

4(a)

any two from:

note that some mark points must have both xylem and phloem mentioned ('v') and some mark points only need xylem or phloem ('or' – below the line)

<i>xylem</i>		<i>phloem</i>	
no cytoplasm / hollow / no contents	v	(peripheral / little / some) cytoplasm I protoplasm R full of cytoplasm / AW or has (a few) organelles A examples of organelles mitochondria or ER I has SER / RER / ribosomes R has nucleus	;
lignified R idea of lignin within element	v	no lignin / (only) cellulose	;
no end wall(s) / no sieve plate(s) A end walls have broken down	or	sieve plate(s) / perforated end wall (s) A (end walls) have sieve pores	
(side walls) contain pits R piths	or	no pits ;	;
no plasmodesmata	or	plasmodesmata (to companion cells) ;	;
thick(er), cell wall / walled	or	thin(ner), cell wall / walled	;

thicker lignified wall = thicker wall mark only (for lignin mark need to state what phloem has)

2

Question	Answer	Marks
4(b)	<i>any two from:</i> to hydrolyse / break, glycosidic bonds ; to produce, reducing sugars / fructose and glucose ; A to make reactive groups (to Benedicts) available to be able to test with Benedict's solution to obtain a positive result ; A description of positive result sucrose, is a non-reducing sugar / has no reactive groups when tested with Benedict's solution ;	2
4(c)(i)	(large number of mitochondria) provide ATP, for active transport / for protein synthesis ; (large number of ribosomes) to synthesise the, proton pumps / carrier proteins / AW ;	2
4(c)(ii)	<i>any three from:</i> H⁺, pumped / actively transported (out of companion cell into cell wall) or <i>ref.</i> to higher concentration of H⁺ (in cell wall) / H⁺ gradient builds up (between cell wall and cytoplasm of companion cell) ; H⁺ move back (into companion cell) by facilitated diffusion ; I sucrose through, <u>co-transporter protein</u> / <u>cotransporter</u> ; A symporter H⁺ cotransports sucrose / sucrose moves (into cell) with H⁺ ; sucrose is transported against the concentration gradient ; A sucrose transported by secondary active transport	3